👩‍🏫 Staff Guide · Small Engines Overhaul · Unit 3 4-Stroke Theory & Engine Parts (Level 2)

Level 2 = about 3rd-grade reading. SLE students stay in the class teacher's lesson and use these pieces next to it: same topics, same order, same points.

1. How the pieces line up

Gen ed piece (Canvas: “Week 2: 4 Stroke Theory PowerPoint (Small Engines Parts ID Reading) + Parts ID quiz, Engine Tools/Parts sheets, Briggs & Stratton disassembly/assembly video, Marine 2-stroke and 4-stroke video notes, iCADEMY Bronze 6 Engine Principles of Operation”)	**Level 2 piece**
4 Stroke Theory PowerPoint (13 slides) + Parts ID quiz	Deck + notes A, B + quiz; Parts ID lab packet
Engine Tools/Parts sheets	Parts ID lab packet + vocab cards
B&S disassembly/assembly video	Video guide (link only)
Marine 2-stroke/4-stroke video notes	Slides “2-Stroke vs 4-Stroke” + notes C
iCADEMY Bronze 6 Engine Principles	Our own slides on the 4 strokes (no STIHL content)
Class slides	Deck: 15 slides, Words to Know first, hold-up-1-2-3 checks, recap
His worksheets (grouped into lessons)	Guided notes: 11 items + key
Class quiz / notes check	Quiz: 10 questions + key
Vocabulary	8 picture cards (cut apart)
—	1 game on the class site: 10 questions + a matching round

2. Supports built in

- Short lines, key words bold, word bank in every notes box
- Quiz: 3 choices, no “all of the above”
- Act out the 4 strokes: students move hands (suck, squeeze, bang, blow)
- Cutaway engine or clear model if the shop has one

3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the “look for” part

4. What to grade

- Parts ID quiz: L1 points to 8 parts on a real engine or picture
- **Notes**: same points as his worksheets for that lesson when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

5. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm in his files say “check with the class teacher”.

🚧 Under construction (check with the class teacher)

- **iCADEMY Bronze 6**: STIHL iCADEMY is login-only; we wrote our own engine-principles content.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.
- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

👩‍🏫 Staff Guide · Engine Parts ID (Level 2)

Small Engines Overhaul · Unit 3 · goes with the class “Small Engine Parts ID Quiz + Engine Tools/Parts sheets” (quiz points). Level 2 = about 3rd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Task card**: Modified from the class “Small Engine Parts ID Quiz + Engine Tools/Parts sheets”. 6 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 3 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- Show me the first step.
- Which tool do we need now?
- Where do your hands go while the tool is turning?
- What should we do when we finish?

3. Modifications

- Task card instead of written directions
- Sentence starters on the reflection
- Partner reads directions aloud

4. What to grade (quiz points)

- Steps done (photo proof)
- Reflection: 2 sentences per box
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver (SLE).

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
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- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

⚙️ 4-Stroke Theory Notes

Small Engines Overhaul · Unit 3 4-Stroke Theory & Engine Parts · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. The strokes

1. Number the strokes in order:

_____ Compression

_____ Exhaust

_____ Intake

_____ Power

2. During the power stroke the _____ plug fires.

Word bank: **spark**

B. Parts

3. Match the part to its job:

Piston	makes spinning motion
Connecting rod	lets gas in or out
Crankshaft	moves up and down
Valve	links piston and crank

C. 2-stroke vs 4-stroke

4. A 2-stroke mixes _____ into the fuel.

5. A 4-stroke makes power every _____ turns.

Word bank: **oil** **two**

Level 2 version of the gen ed “4 Stroke Theory PowerPoint + Parts ID quiz.”

📖 4-Stroke Theory & Engine Parts: Vocabulary Picture Cards (Level 2)

Name: _____

Cut on the dashed lines. Use the cards to match, sort, or quiz a partner.



Intake

Stroke 1: the piston moves down and the intake valve opens to pull in the air-fuel mix.



Compression

Stroke 2: the piston moves up with both valves closed and squeezes the mix.



Power

Stroke 3: the spark plug fires, the mix burns and pushes the piston down.



Exhaust

Stroke 4: the piston moves up and the exhaust valve opens to push burned gas out.



Piston

The round part that slides up and down in the cylinder.



Crankshaft

The shaft that changes the piston's up-and-down motion into spinning motion.



Valve

A part that opens and closes to let the mix in or exhaust out.



Flywheel

A heavy wheel on the crankshaft that keeps the engine turning smoothly and holds the magnets for the spark.

🔩 Engine Parts ID

Small Engines Overhaul · Unit 3 · Level 2



Name: _____

Modified from the class “Small Engine Parts ID Quiz + Engine Tools/Parts sheets”.

🧺 Supplies

- ☐ ⚙️ **Practice engine** staff choose
- ☐ 🏷️ **Part label cards**
- ☐ 🧰 **Tool tray**
- ☐ 📋 **Parts ID sheet**
- ☐ 📸 **Camera/iPad**

🔩 Steps: check each box when you finish

1	🔌	Check the spark plug wire is disconnected.	☐
2	🏷️	Label 8 outside parts: spark plug, air filter, carburetor, flywheel, muffler, fuel tank, recoil starter, governor.	☐
3	🔧	Match each part to the tool that removes it.	☐
4	📋	Write each part's job on the sheet.	☐
5	⚙️	With adult supervision, turn the flywheel slowly and watch the valves (if visible).	☐
6	📸	Photo of your labeled engine.	☐

🔩 Engine Parts ID · Prep Sheet

Small Engines Overhaul · Unit 3 · Level 2



Name: _____

👥 Job roles: circle your job. Initial when it is done.

Job	What you do	Done ✔️
🔧 **Mechanic**	Does the main engine job with the tools	
🧰 **Tool helper**	Gets tools from the board and returns them to staff	
📋 **Recorder**	Takes photos and fills in the chart	
🧽 **Clean-up captain**	Wipes the bench, sweeps, checks tools are back	

🧰 Tools and supplies: check when you have it

- ☐ ⚙️ **Practice engine** staff choose
- ☐ 🧰 **Tool tray**
- ☐ 📸 **Camera/iPad**
- ☐ 🏷️ **Part label cards**
- ☐ 📋 **Parts ID sheet**

⚠️ Safety and manners focus

🥽 **Safety glasses** on the whole time.

👖 **Coveralls** and closed-toe shoes; tie back long hair.

🔧 Use tools **with adult supervision**. Right tool for the job.

🔌 **Spark plug wire off** before touching the engine.

🧼 **Clean up** and wash hands when you finish.

✅ Before you start

☐ Safety glasses ☐ Coveralls ☐ Adult at bench

🔩 Engine Parts ID · Reflection

Small Engines Overhaul · Unit 3 · Level 2



Name: _____

📸 My photos / proof

☐ Before photo ☐ During photo ☐ Finished photo

✅ Skills I used today (check all you did)

☐ 🥽 Wore safety glasses and used tools safely

☐ 🔧 Finished the Engine Parts ID steps

☐ 🧽 Cleaned up and returned tools

☐ 🤝 Did my job role

What did you do today, in order?

1. _____
2. _____

Ideas: **First…** **Next…** **Last…**

What did you learn about engines?

1. _____
2. _____

Ideas: **The engine needs…** **I saw…** **Next time…**

✅ 4-Stroke Theory & Engine Parts Quiz

Small Engines Overhaul · Unit 3 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. ⬇️ The first stroke is…

A. Power | B. Intake | C. Exhaust

2. 🗜️ During compression the valves are…

A. Open | B. Missing | C. Closed

3. 💥 The stroke that pushes the piston down with burning fuel is…

A. Power | B. Intake | C. Exhaust

4. 💨 Burned gas leaves on the…

A. Intake stroke | B. Compression stroke | C. Exhaust stroke

5. 🔗 The connecting rod links the piston to the…

A. Crankshaft | B. Muffler | C. Air filter

6. 🔁 The crankshaft turns how many times per 4 strokes?

A. 4 | B. 2 | C. 1

7. 🚪 Valves are opened by the…

A. Gas tank | B. Camshaft | C. Flywheel

8. 🔘 The flywheel…

A. Keeps the engine turning smoothly | B. Holds the gas | C. Cleans the air

9. 2️⃣ A 2-stroke engine gets its oil…

A. From the crankcase only | B. From water | C. Mixed in the fuel

10. 4️⃣ A lawn mower usually has a…

A. 4-stroke engine | B. Jet engine | C. Electric motor only

🎬 Engine Disassembly / Assembly Video

Small Engines Overhaul · Unit 3 · Level 2 video guide (modified). Briggs & Stratton video linked in Canvas (YouTube). Link only.

Name: _____

Before you watch

Name one part you think you will see. _____

While you watch: circle the best answer

1. What comes off first?

A. The piston | B. Outside parts: shroud, air filter, carburetor | C. The crankshaft

2. What tool removes the flywheel?

A. A hammer only | B. Pliers | C. A flywheel puller

3. Where do small parts go?

A. In a labeled tray | B. On the floor | C. In a pocket

4. Reassembly is done…

A. Randomly | B. Without a manual | C. In reverse order

After you watch: finish the sentence

5. After watching, finish the sentence:

Taking photos during teardown helps because

_____

Goes with the class “Engine Disassembly/Assembly Video” (class points pts). Same points when complete.

⚙️ 4-Stroke Theory Notes · ANSWER KEY

Small Engines Overhaul · Unit 3 4-Stroke Theory & Engine Parts · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. The strokes

1. Number the strokes in order:

2 Compression

4 Exhaust

1 Intake

3 Power

2. During the power stroke the **spark** plug fires.

Word bank: **spark**

B. Parts

3. Match the part to its job:

(Key: Piston → moves up and down · Connecting rod → links piston and crank · Crankshaft → makes spinning motion · Valve → lets gas in or out)

Piston	makes spinning motion
Connecting rod	lets gas in or out
Crankshaft	moves up and down
Valve	links piston and crank

C. 2-stroke vs 4-stroke

4. A 2-stroke mixes **oil** into the fuel.

5. A 4-stroke makes power every **two** turns.

Word bank: **oil** **two**

Level 2 version of the gen ed "4 Stroke Theory PowerPoint + Parts ID quiz."
🔑 Key written by our team from the class teacher's materials, please check.

✅ 4-Stroke Theory & Engine Parts Quiz · ANSWER KEY

Small Engines Overhaul · Unit 3 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. ⬇️ The first stroke is… Parts ID

A. Power | **B. Intake** | C. Exhaust

2. 🗜️ During compression the valves are… Parts ID

A. Open | B. Missing | **C. Closed**

3. 💥 The stroke that pushes the piston down with burning fuel is… Parts ID

A. Power | B. Intake | C. Exhaust

4. 💨 Burned gas leaves on the… Parts ID

A. Intake stroke | B. Compression stroke | **C. Exhaust stroke**

5. 🔗 The connecting rod links the piston to the… Parts ID

A. Crankshaft | B. Muffler | C. Air filter

6. 🔁 The crankshaft turns how many times per 4 strokes? Parts ID

A. 4 | **B. 2** | C. 1

7. 🚪 Valves are opened by the… Parts ID

A. Gas tank | **B. Camshaft** | C. Flywheel

8. 🔘 The flywheel… Parts ID

A. Keeps the engine turning smoothly | B. Holds the gas | C. Cleans the air

9. 2️⃣ A 2-stroke engine gets its oil… Marine video

A. From the crankcase only | B. From water | **C. Mixed in the fuel**

10. 4️⃣ A lawn mower usually has a… Marine video

A. 4-stroke engine | B. Jet engine | C. Electric motor only

Answers

1-B · 2-C · 3-A · 4-C · 5-A · 6-B · 7-B · 8-A · 9-C · 10-A

Reworded from the gen ed “Small Engine Parts ID Quiz”. Score: __ / 10. 🔑 Key written by our team from Mr. Davis's materials, please check.

🎬 Engine Disassembly / Assembly Video · ANSWER KEY

Small Engines Overhaul · Unit 3 · Level 2 video guide (modified). Briggs & Stratton video linked in Canvas (YouTube). Link only.

Name: _____

Before you watch

Name one part you think you will see. _____

While you watch: circle the best answer

1. What comes off first?

A. The piston | **B. Outside parts: shroud, air filter, carburetor** | C. The crankshaft

2. What tool removes the flywheel?

A. A hammer only | B. Pliers | **C. A flywheel puller**

3. Where do small parts go?

A. In a labeled tray | B. On the floor | C. In a pocket

4. Reassembly is done…

A. Randomly | B. Without a manual | **C. In reverse order**

After you watch: finish the sentence

5. After watching, finish the sentence:

Taking photos during teardown helps because **you can see how parts go back together**

Goes with the class “Engine Disassembly/Assembly Video” (class points pts). Same points when complete.
🔑 Key written by our team from the class teacher's materials, please check.